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A systematic survey of artificial intelligence methods for ECG-based cardiovascular disease prediction

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Abstract

Background Cardiovascular diseases continue to be the leading cause of death globally, creating an urgent need for accurate and scalable diagnostic approaches. The electrocardiogram remains essential for cardiac assessment, yet the growing demand for expert interpretation places increasing strain on healthcare systems and introduces potential diagnostic variability.

Objective This systematic review examines how artificial intelligence methods are being applied to ECG-based cardiovascular disease prediction, with particular attention to data handling practices, modeling approaches, interpretability techniques, and evaluation strategies. The review aims to provide researchers with a comprehensive understanding of the field's current state while identifying persistent challenges and opportunities for meaningful progress.

Methods Following PRISMA guidelines, we systematically searched PubMed, IEEE Xplore, Scopus, Web of Science, and other major databases for peer-reviewed studies published between January 2020 and November 2025. After screening 1247 records, 83 high-quality studies met our inclusion criteria and underwent rigorous quality assessment using a customized evaluation tool addressing data sources, preprocessing, model development, and clinical applicability.

Results The analysis reveals that while deep learning approaches—particularly convolutional neural networks and hybrid architectures—have demonstrated impressive performance in controlled settings, significant gaps remain. Only 38.6% of studies showed low risk of bias, with external validation lacking in 72% of cases. Dataset diversity remains problematic, with 78% of data originating from North America or Europe and minimal representation of pediatric populations. Explainable AI methods appear in only 52% of recent studies, and demographic fairness assessments remain rare.

Conclusion This review provides a structured roadmap for researchers navigating the entire AI-ECG pipeline, from data acquisition through model deployment. By systematically identifying methodological gaps and proposing concrete solutions, we aim to accelerate the development of reliable, fair, and clinically useful AI systems for cardiovascular care.

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Keywords Cardiovascular disease, Electrocardiogram, Machine learning, Deep learning, Explainable artificial intelligence, Clinical validation

1 Introduction

1.1 The global burden of cardiovascular disease

Cardiovascular diseases represent one of the most formidable challenges facing modern medicine. These conditions—including coronary artery disease, heart failure, stroke, and various rhythm disorders—affect hundreds of millions of people worldwide and claim nearly 18 million lives each year [1]. Beyond these stark mortality statistics lies an equally concerning reality: cardiovascular disease profoundly diminishes quality of life through symptoms like persistent chest discomfort, debilitating shortness of breath, and progressive physical limitation.

Despite decades of advances in prevention, diagnosis, and treatment, cardiovascular disease remains stubbornly prevalent. The reasons for this persistence are multifaceted, reflecting the complex interplay of genetic predisposition, lifestyle factors, aging populations, and healthcare access disparities. What makes cardiovascular disease particularly challenging is its heterogeneity—patients present with diverse symptoms, varying disease stages, and often multiple concurrent conditions that complicate diagnosis and management.

1.2 The electrocardiogram: a window into cardiac health

Since its introduction into clinical practice over a century ago, the electrocardiogram has served as an indispensable tool for understanding cardiac function [2]. By capturing the heart's electrical activity through electrodes placed on the skin, the ECG provides a real-time representation of cardiac depolarization and repolarization. The characteristic waveform—comprising the P wave (atrial depolarization), QRS complex (ventricular depolarization), and T wave (ventricular repolarization)—offers clinicians a non-invasive window into cardiac physiology (Fig. 1).

When physicians examine an ECG tracing, they look for subtle clues hidden within these waveforms. A slightly elevated ST segment might signal an evolving myocardial infarction. An irregularly irregular rhythm could indicate atrial fibrillation. A prolonged QT interval might suggest susceptibility to dangerous arrhythmias [3]. These pattern recognition tasks, while routine for experienced cardiologists, require years of training and remain subject to inter-observer variability.

1.3 The evolution of computer-assisted ECG interpretation

The dream of automating ECG interpretation is nearly as old as the ECG itself. Researchers began developing computer-assisted approaches more than fifty years ago, initially focusing on rule-based systems that measured predefined features and compared them against established criteria [4]. These early systems, while groundbreaking for their time, suffered from significant limitations. Their accuracy plateaued at levels below expert human interpretation, they struggled with noisy or atypical recordings, and they could not adapt to evolving clinical knowledge.

The fundamental limitation of these traditional approaches lay in their reliance on hand-crafted features. Engineers and clinicians had to decide in advance which aspects

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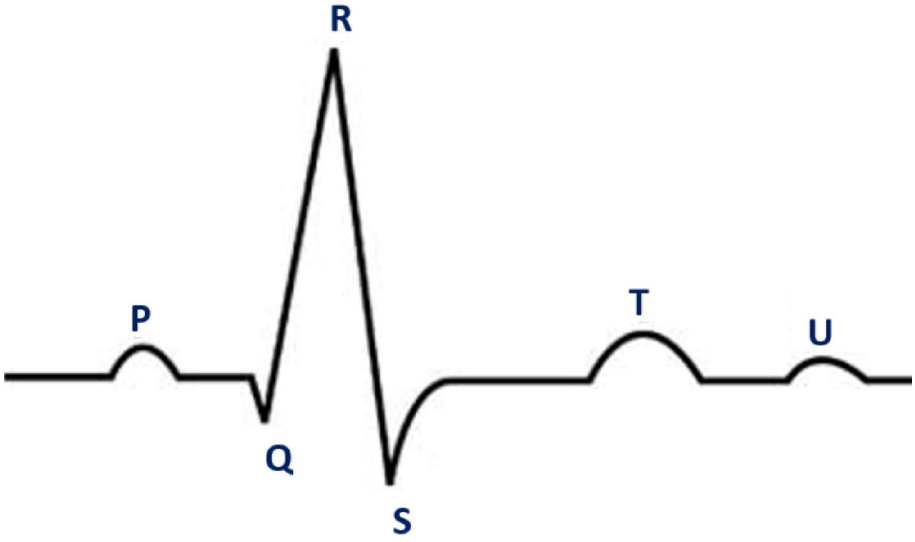


Fig. 1 Components of a standard ECG waveform

of the ECG signal mattered—amplitudes here, intervals there—and program these measurements into rigid decision rules. This approach could not capture the nuanced, holistic pattern recognition that expert interpreters develop through years of experience.

1.4 The promise of artificial intelligence

Th emergence of artificial intelligence, particularly machine learning and deep learning, has fundamentally altered this landscape [5]. Rather than relying on predetermined features, AI methods can learn directly from data, discovering patterns that may escape even expert observation. A convolutional neural network trained on thousands of ECG tracings might learn to recognize subtle morphological variations that precede ventricular tachycardia. A long short-term memory network might capture temporal dependencies spanning dozens of heartbeats that signal paroxysmal atrial fibrillation.

Recent clinical investigations have demonstrated that AI models can match or exceed human performance on specific diagnostic tasks [6]. When combining ECG analysis with clinical data, these models have shown particular promise for predicting future cardiovascular events—sometimes identifying patients at risk months before symptoms appear. Yet for all this promise, the translation of AI systems from research laboratories to clinical practice has proceeded slowly.

1.5 Barriers to clinical adoption

The gap between research promise and clinical reality reflects several interconnected challenges. First, AI models, particularly deep learning approaches, often function as "black boxes"—they produce accurate predictions but offer little insight into their reasoning [7]. A physician who receives an alert that a patient faces high risk of heart failure may reasonably ask: "Why this patient? What features drove this prediction? Should I trust this output?" Without answers to these questions, many clinicians remain justifiably hesitant to incorporate AI recommendations into their decision-making.

Second, the datasets used to train AI models often fail to reflect the diversity of real-world clinical populations [6]. A model trained primarily on middle-aged European men may perform poorly when deployed in a diverse urban hospital serving patients of varied

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ages, ethnicities, and socioeconomic backgrounds. Such performance degradation—known as dataset shift—raises serious concerns about fairness and equity in AI-assisted care.

Third, most published studies prioritize a single metric—accuracy—while paying insufficient attention to the factors that determine real-world utility: robustness to noise, interpretability, computational efficiency, and seamless integration into clinical workflows [8].

1.6 The emergence of explainable AI

Recognizing these limitations, researchers have increasingly turned to explainable artificial intelligence (XAI) methods [8]. These techniques aim to open the black box, providing insights into how models arrive at their predictions. Some approaches generate feature importance scores, highlighting which aspects of the ECG signal most influenced a particular decision. Others produce visual explanations, creating saliency maps that spotlight critical regions of the waveform. Still others build inherently interpretable models that trade some predictive power for transparency.

The promise of XAI extends beyond satisfying scientific curiosity. When clinicians understand why a model makes specific recommendations, they can exercise informed judgment—agreeing when the reasoning aligns with their clinical intuition, questioning when it does not, and learning when the model identifies patterns they might have missed. This collaborative paradigm, where human expertise and machine intelligence complement each other, offers the most realistic path toward meaningful clinical integration.

1.7 Why this review matters now

Several recent surveys have examined AI applications in ECG analysis, yet significant gaps remain in the literature. Some reviews focus narrowly on specific conditions like arrhythmia while neglecting broader cardiovascular applications. Others provide comprehensive coverage but lack rigorous quality assessment of included studies. Many fail to address critical issues like dataset diversity, model fairness, and clinical translation barriers.

The rapid pace of methodological development compounds these limitations. The landscape of AI-ECG research has transformed dramatically over the past five years, with transformer architectures, foundation models, and advanced XAI techniques entering the mainstream. A review that captures these developments while maintaining rigorous methodological standards serves an urgent need.

1.8 What this review offers

This systematic review aims to provide researchers—whether new to the field or seeking to deepen their expertise—with a comprehensive, critically evaluated roadmap for AI-based ECG analysis. We examine the entire pipeline:

- *Data acquisition* What datasets exist, how representative are they, and what are their limitations?
- *Preprocessing* How do researchers clean and prepare ECG signals, and what trade-offs do different approaches entail?

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- *Model development* Which machine learning and deep learning architectures show greatest promise for different clinical tasks?
- *Explainability* How can researchers make model decisions transparent and clinically meaningful?
- *Evaluation* What metrics best capture model performance, and how should researchers assess generalizability?
- *Clinical translation* What barriers separate research prototypes from deployed clinical tools?

1.9 Scope and organization

We systematically reviewed 83 high-quality studies published between January 2020 and November 2025, selected through rigorous screening and quality assessment. Throughout this review, we emphasize critical analysis over mere description—identifying not just what researchers have done, but what works, what doesn't, and where the field should focus its efforts.

The remainder of this paper proceeds as follows. Section 2 details our systematic review methodology, including search strategies, inclusion criteria, and quality assessment procedures. Section 3 examines data acquisition practices and critically analyzes dataset characteristics. Section 4 explores preprocessing techniques and their implications for model performance. Section 5 reviews machine learning approaches, while Sect. 6 examines deep learning architectures. Section 7 investigates explainable AI methods and their role in clinical adoption. Section 8 discusses evaluation metrics and their interpretation. Section 9 synthesizes identified challenges and proposes future research directions. Section 10 offers concluding reflections on the state and trajectory of the field.

2 CVD detection methodology

2.1 Research questions

To guide this systematic review, five questions were formulated research questions that span the entire AI development pipeline:

Table 1 outlines the key research questions guiding the review of ECG-based cardiovascular analysis using artificial intelligence. Each question highlights a specific focus area, ranging from data acquisition and model development to evaluation methods, interpretability, and future research challenges. Together, these questions structure the investigation and help identify important trends and gaps in current studies.

Table 1 Research questions and focus areas for ECG-based cardiovascular disease analysis

Research question	Focus area
RQ1: What workflows are commonly used for collecting, curating, and preprocessing ECG data in CVD detection, diagnosis, prognosis, and monitoring studies?	Data acquisition and preparation
RQ2: Which machine learning and deep learning algorithms have been applied to ECG-based cardiovascular analysis across different clinical tasks?	Model development
RQ3: What clinical performance measures are typically used to assess the utility, reliability, and robustness of AI models for ECG-based cardiovascular applications?	Evaluation methodologies
RQ4: Which explainable AI approaches have been integrated into ECG-driven systems, and how do they affect model interpretability, transparency, and potential for clinician trust?	Interpretability
RQ5: What challenges, limitations, ethical considerations, and future research directions have been identified in the literature?	Knowledge gaps

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2.2 Search strategy

Databases Searched We conducted literature searches across five major databases: Google Scholar, Wiley Online Library, PubMed, ScienceDirect, and SpringerLink.

Search Period January 2020 to November 2025.

Search Strings The following Boolean search strings were used, adapted for each database's syntax:

- "Artificial Intelligence" AND "Electrocardiogram" AND "Cardiovascular Disease"
- "Deep Learning" AND "ECG" AND "Heart Disease"
- "Machine Learning" AND "Electrocardiography" AND "Classification"
- "Neural Networks" AND "Cardiac Arrhythmia" AND "Prediction"

2.3 Inclusion and exclusion criteria

Table 2 summarizes the criteria used to determine which studies were included or excluded during the literature screening process. The guidelines ensure that only recent, peer-reviewed research focusing on AI-based ECG analysis for cardiovascular diseases is considered.

2.4 Study selection process

Two independent reviewers (TM and SMP) screened all identified records. Disagreements were resolved through discussion or consultation with a third reviewer (PLG). The selection process followed PRISMA 2020 guidelines as illustrated in Fig. 2.

2.5 Quality assessment

Study quality was assessed using a customized tool addressing: data sources, preprocessing clarity, model development rigor, performance evaluation, external validation, reproducibility, and interpretability. Studies were categorized as low, moderate, or high risk of bias.

Results Among 83 included studies:

- Low risk of bias: 32 (38.6%)
- Moderate risk of bias: 41 (49.4%)
- High risk of bias: 10 (12.0%)

2.6 Review scope and roadmap

This review examines the complete workflow of AI-based CVD detection using ECG signals—from data collection through model evaluation—as illustrated in Fig. 3. Each subsequent section identifies methodological gaps that influence the reliability and generalizability of current approaches.

Table 2 Study inclusion and exclusion criteria for literature selection

Criterion	Inclusion	Exclusion
Publication type	Peer-reviewed journal articles, conference proceedings	Preprints, editorials, opinion pieces
Time frame	January 2020 – November 2025	Before 2020
Language	English	Non-English
Topic	AI/ML/DL applied to ECG for CVD	Non-ECG studies, non-AI methods
Outcome reporting	Quantitative performance metrics	Qualitative only

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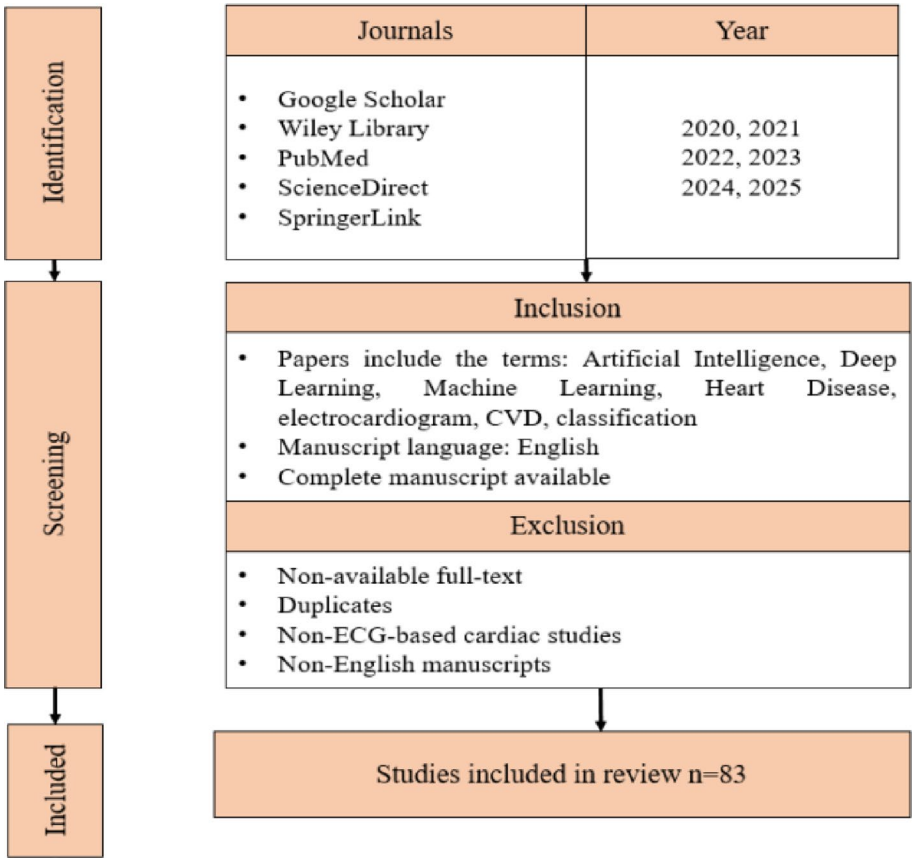


Fig. 2 Literature selection process following PRISMA method

3 Data acquisition

The effectiveness and generalizability of AI-based cardiovascular disease prediction models depend fundamentally on the quality and characteristics of the ECG data used for training and evaluation. Our review reveals a pronounced reliance on publicly accessible ECG repositories, with limited use of institution-specific or custom-collected datasets. Among the 83 studies analyzed, 95% utilized public data sources, 3% employed private or institutionally restricted clinical data, and 2% involved custom-collected datasets.

3.1 Private clinical datasets

Only 3% of reviewed studies incorporated private or institutionally regulated ECG datasets. These datasets are typically obtained from cardiology departments, hospital information systems, and specialized diagnostic laboratories. While they offer high-fidelity signals and detailed clinical annotations, their use is constrained by ethical considerations, institutional policies, and patient confidentiality requirements. Access is generally limited to researchers with direct clinical collaborations. Despite their small scale, these datasets capture real-world clinical variability that is often underrepresented in public repositories.

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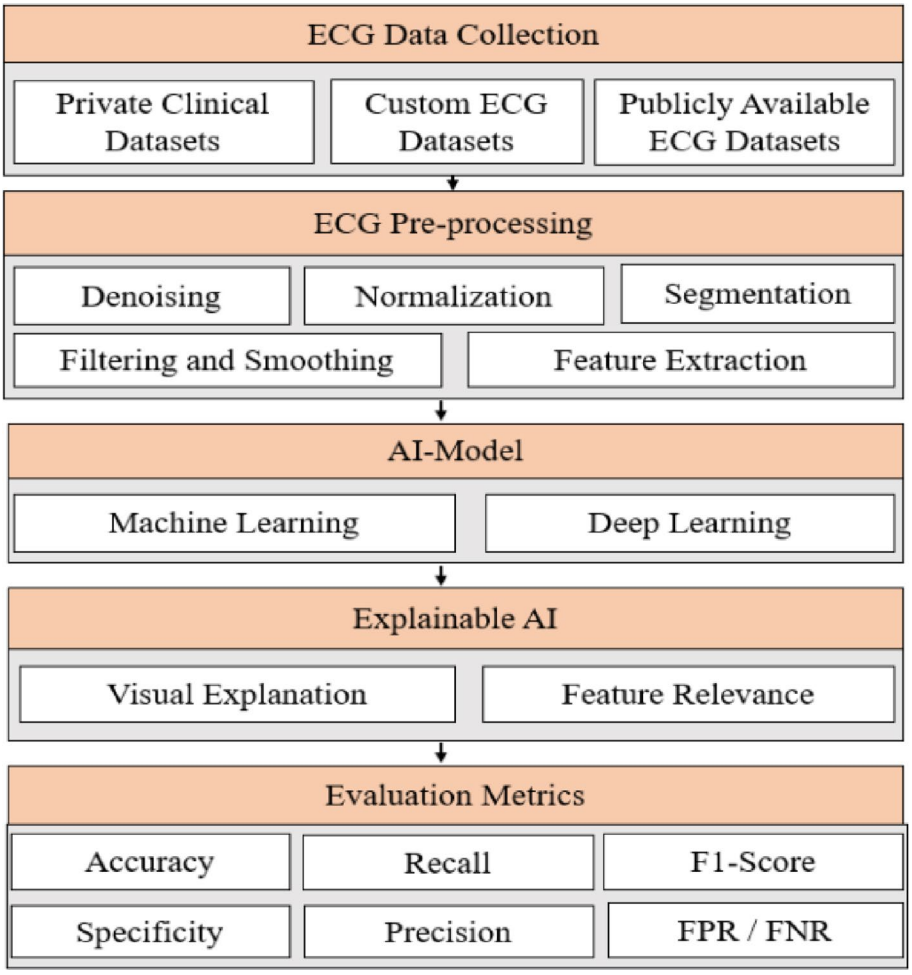


Fig. 3 Workflow of AI-based CVD detection using ECG signals

3.2 Custom ECG datasets

Approximately 2% of studies involved researchers collecting their own ECG data. This approach is uncommon due to substantial costs, ethical approval requirements, and the need for controlled acquisition protocols. Custom datasets are typically generated using wearable recorders, portable devices, or specialized prototypes tailored to specific research objectives. Recordings may be obtained under resting, ambulatory, exercise, or continuous monitoring conditions, enabling creation of highly focused datasets aligned with real-world applications. Signals are typically annotated by cardiologists or trained technicians, sometimes with automated assistance followed by manual verification. These datasets, while small, provide accurate and contextually rich information.

3.3 Publicly available ECG datasets

The vast majority of studies (95%) relied on publicly available ECG repositories. These open datasets offer standardized signal formats, expert annotations, and broad diagnostic categories, making them well-suited for training and benchmarking AI models. Their accessibility facilitates reproducibility and comparative evaluation, explaining their widespread adoption. Table 3 summarizes the most commonly used public ECG datasets in CVD prediction research.

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Table 3 Summary of public ECG datasets

Database	Device	Fs (Hz)	Dur	Rec	Classes
MIT-BIH Arrhythmia [9]	Amb. ECG	360	30 m	48	N, RBBB, LBBB, SVTA, PAC, AFIB, PVC, VFIB, AFL, J, AB, VB, BII, SBR, VT, VFL, WPW, IVR
MIT-BIH AF [10]	Amb. ECG	250	600 m	25	AFIB, AFL, J, Others
MIT-BIH NSR [11]	N/S	128	1500 m	18	N
PTB Diagnostic [12]	PTB Rec	1000	Var	549	N, VALV, BBB, MI, HF, CM, MC, DYS, MH, MISC
PTB-XL [13]	Schiller	500	10 s	21,799	N, MI, ST-T, CD, HYP
PhysioNet CinC 2017 [14]	AliveCor	300	0.5–1 m	12,186	N, AFIB, Noisy, Other
MIT-BIH SV Arr. [15]	N/S	128	30 m	78	SVTA
St Petersburg INCART [16]	Holter	257	30 m	75	MI, WPW, SVE, HTN, AFIB, SVTA, AVB, BBB
BIDMC CHF [17]	Amb. ECG	250	1200 m	15	CHF
China Challenge 2018 [18]	N/S	500	6–60 s	9,831	N, I-AVB, AFIB, RBBB, LBBB, PVC, PAC, ST-E, ST-D
Long-Term AF [19]	N/S	128	1440–1500 m	84	Normal, PAC, PVC
Georgia 12-Lead [20]	N/S	500	10 s	20,678	N, AFL, AVB, Brady, AFIB, RBBB, LBBB, PAC, PVC
Long-Term ST [21]	N/S	250	1260–1440 m	86	ST changes
MIT-BIH Malignant V. [22]	N/S	360	30 m	22	VT, VFIB, VFL
European ST-T [23]	Amb. ECG	250	120 m	90	MI, ST
Shaoxing-Ningbo [24]	GE	500	10 s	10,646	N, STC, SVTA, AFIB, AVNRT, ATC
MIT-BIH Long-Term [25]	N/S	128	840–1500 m	7	Normal, PAC, PVC

AFIB, atrial fibrillation; AFL, atrial flutter; ATC, atrial tachycardia; AVB, atrioventricular block; AVNRT, AV node reentrant tachycardia; AVRT, AV reentrant tachycardia; BBB, bundle branch block; Brady, bradycardia; CD, conduction disturbance; CHF, congestive heart failure; CM, cardiomyopathy; HF, heart failure; HYP, hypertrophy; MI, myocardial infarction; N, normal sinus rhythm; PAC, premature atrial contraction; PVC, premature ventricular contraction; RBBB, right bundle branch block; LBBB, left bundle branch block; SBR, sinus bradycardia; SVTA, supraventricular tachyarrhythmia; ST-E, ST elevation; ST-D, ST depression; VFIB, ventricular fibrillation; VT, ventricular tachycardia; WPW, Wolff–Parkinson–White syndrome

3.4 Critical analysis of dataset limitations

Demographic Diversity Analysis reveals that 78% of public datasets originate from North America or Europe. Ethnicity data is reported in fewer than 10% of datasets, and pediatric populations are severely underrepresented (less than 3% of recordings). Average female representation across datasets is 41.5%.

Annotation Quality Only 35% of datasets specify annotator qualifications. Inter-annotator agreement metrics are reported for just 22% of datasets. Standardized annotation guidelines are described in only 40% of cases.

Generalizability Implications Studies reporting cross-dataset validation show average performance degradation of 8–15% when models are tested on data from different sources. Only 28% of studies include any form of external validation.

4 Data preprocessing and signal conditioning

Raw ECG signals acquired from clinical or wearable devices are frequently contaminated with various forms of noise and artifacts that can compromise subsequent analysis [26]. These distortions arise from multiple sources: patient movement, electrode displacement, respiratory activity, muscle contractions, and powerline interference. Without appropriate preprocessing, even sophisticated AI models may struggle to extract reliable diagnostic information. Effective preprocessing pipelines are therefore essential for ensuring data integrity, consistency, and clinical relevance before downstream analysis.

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4.1 Denoising

Denoising techniques target specific types of signal corruption including baseline wander, electromyographic interference, powerline noise, and motion artifacts. Traditional approaches employ frequency-selective filters such as low-pass, high-pass, and notch filters to remove unwanted frequency components while preserving the diagnostic ECG spectrum.

More advanced methods have emerged to address the non-stationary nature of ECG noise. Wavelet transform-based denoising offers time–frequency decomposition that can isolate noise while preserving transient signal features [27]. Empirical mode decomposition (EMD) adaptively decomposes signals into intrinsic mode functions, enabling selective noise removal [27]. Adaptive filtering techniques[28] including Least Mean Squares (LMS) and Normalized LMS algorithms dynamically adjust filter parameters based on signal characteristics [29]. Kalman filtering provides recursive estimation suitable for real-time applications.

Recent innovations include neural network-based denoising using autoencoders and generative adversarial networks (GANs), which learn noise patterns directly from data [29]. Hybrid approaches combining classical filters with machine learning models have shown promise for balancing noise reduction with signal preservation.

4.2 Filtering and smoothing

Filtering operations further refine denoised signals by isolating the ECG frequency band of interest. Bandpass filters with typical cutoffs between 0.5 and 150 Hz remove both low-frequency baseline drift and high-frequency interference while preserving the diagnostically relevant spectrum [30]. Notch filters specifically target powerline interference at 50 or 60 Hz depending on regional electrical standards [30]. Butterworth filters are widely employed due to their maximally flat frequency response [31].

Smoothing techniques reduce transient spikes and high-frequency fluctuations. Moving average filters and median filters offer simple yet effective smoothing, while more sophisticated approaches include Savitzky-Golay filters that preserve higher-order moments, smoothing splines that balance fit and smoothness, and wavelet-based algorithms that simultaneously denoise and smooth while maintaining morphological fidelity.

4.3 Segmentation

Segmentation isolates individual cardiac cycles or specific waveform components to facilitate feature extraction and disease classification [32]. Traditional approaches include threshold-based methods and slope-detection algorithms, notably the Pan-Tompkins technique for QRS complex identification. Template matching aligns heartbeats against representative templates [33], while Hidden Markov Models capture sequential dependencies in ECG morphology [34]. Wavelet-based multiresolution analysis enables detection across multiple time scales.

Contemporary methods increasingly employ deep learning, particularly convolutional neural networks and U-Net architectures, which demonstrate robust segmentation even on noisy or morphologically abnormal recordings. These data-driven approaches learn to identify relevant boundaries directly from labelled examples.

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4.4 Normalization

Normalization reduces inter- and intra-subject variability by standardizing beat characteristics. R-peak detection typically serves as the reference point for heartbeat alignment. Time normalization resamples cardiac cycles to uniform lengths, enabling consistent comparison across recordings with varying heart rates. Amplitude normalization scales signals to consistent ranges, mitigating variability arising from differences in electrode placement, skin impedance, or recording equipment. These standardization steps improve dataset comparability and enhance the robustness of downstream machine learning models.

4.5 Feature extraction

Feature extraction transforms preprocessed signals into representations suitable for classification. Traditional approaches compute morphological and temporal features including waveform amplitudes, interval durations (PR, QRS, QT), slopes, and angles. Accurate preprocessing is critical to ensure that extracted features reflect true physiological signals rather than noise artifacts.

Contemporary deep learning methods increasingly bypass explicit feature extraction by learning representations directly from data. However, these approaches remain heavily dependent on high-quality pre-processed inputs. Table 4 summarizes recent work on ECG preprocessing, highlighting key insights and identified research gaps.

5 Machine learning approaches

Machine learning methods for ECG-based cardiovascular disease prediction fall into two broad categories: traditional approaches requiring explicit feature extraction and deep learning methods that learn representations directly from data. This section

Table 4 Overview of existing ECG preprocessing methods and identified limitations

Ref	Preprocessing technique	Key insight	Research gap
[35]	Review: Wavelet, EMD/CEEMD, adaptive & notch filters	Hybrid pipelines improve mixed-noise ECG denoising	No standard benchmarks for morphology preservation; limited real-world validation
[36]	EMD-based adaptive noise removal	Effective baseline wander removal with QRS preservation	Mode mixing, edge effects, high computational cost
[37]	FIR/IIR, median, wavelet, EMD, adaptive filter comparison	Adaptive filters reduce motion artifacts; wavelets remove HF noise	Poor handling of non-stationary multi-source noise
[38]	Hybrid artifact suppression algorithm	Improves ambulatory ECG quality for arrhythmia detection	Performance drops when motion overlaps ECG bands
[39]	CEEMD + CNN classification	Better mode separation and diagnostic accuracy	High computation; real-time and cross-dataset limits
[40]	Motion artifact reduction for wearable ECG	Motion modeling improves QRS preservation	Requires motion context or extra sensors
[41]	Wavelet denoising with optimized thresholds	Higher SNR and morphology preservation	Dataset-dependent threshold tuning
[42]	Adaptive notch filter with metaheuristic tuning	Improved powerline interference suppression	High computational overhead
[43]	Adaptive noise cancellation	Significant SNR improvement using reference noise	Requires reliable noise reference
[44]	Adaptive filtering for AF detection	Improves classification accuracy	Pathology-specific preprocessing limits generality
[45]	Fully-Gated Denoising Autoencoder (FGDAE)	Learns complex overlapping noise patterns	Needs large labeled datasets; interpretability issues

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examines conventional ML techniques identified in our review, while Section VI covers deep learning architectures.

5.1 Overview of traditional ML methods in ECG analysis

Among the 83 reviewed studies, 41 (49.4%) employed traditional machine learning approaches, either as primary classifiers or as benchmarks against deep learning methods. These techniques share a common workflow: hand-crafted features extracted from preprocessed ECG signals serve as inputs to classification algorithms. Feature sets typically include morphological characteristics (amplitudes, intervals, slopes), statistical measures (mean, variance, entropy), and frequency-domain representations.

5.2 Commonly used ML classifiers

Table 5 presents the major machine learning algorithms commonly applied in ECG-based cardiovascular disease studies along with their typical usage frequency. It highlights their primary clinical applications, strengths, and practical limitations. The comparison provides insight into how different models contribute to diagnosis and prediction tasks in cardiovascular research.

5.3 Comparative performance insights

Accuracy Comparisons Across studies reporting direct comparisons on identical datasets, ensemble methods (particularly Random Forest and Gradient Boosting) consistently outperform single classifiers by 4–7% in accuracy. However, all traditional ML approaches underperform compared to deep learning models when applied to raw or minimally preprocessed ECG signals, with performance gaps of 8–15% reported in benchmark comparisons [46, 47].

Feature Engineering Impact The quality of hand-crafted features significantly influences ML performance. Studies employing expert-derived features (based on clinical knowledge of ECG morphology) achieve 5–10% higher accuracy than those using generic statistical features. However, this dependence on expert knowledge limits scalability and may miss novel patterns detectable by deep learning [48].

Interpretability Advantage Despite lower raw accuracy, traditional ML methods offer greater inherent interpretability. Feature importance scores from Random Forest models

Table 5 Summary of machine learning methods in ECG-based CVD detection

Method	Usage (%)	Typical application	Key advantage	Limitation
Support Vector Machine (SVM)	23%	Arrhythmia classification, MI detection	Effective in high-dimensional space; strong margins	Kernel selection sensitive; high computational cost
Random Forest (RF)	18%	Multi-class classification, feature ranking	Handles non-linear data; reduces overfitting	Large models reduce interpretability
Logistic Regression (LR)	12%	Binary diagnosis, risk prediction	Interpretable; fast training; probabilistic output	Cannot model non-linear relations
K-Nearest Neighbors (KNN)	8%	Pattern matching, small datasets	Simple; no training phase	Slow inference; dimensionality issues
Decision Tree (DT)	9%	Rule-based diagnosis	Highly interpretable; handles mixed data	Overfitting; unstable structure
Naïve Bayes (NB)	5%	Baseline models, real-time prediction	Very fast; works with small datasets	Independence assumption unrealistic
Ensemble Methods	25%	Performance improvement	Higher accuracy; variance reduction	Increased complexity; lower interpretability

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and coefficient magnitudes from logistic regression provide direct insights into which ECG characteristics drive predictions—a crucial advantage for clinical adoption [49].

Sample Efficiency Traditional ML methods require substantially less training data than deep learning approaches. Studies with fewer than 1,000 recordings predominantly use ML methods, achieving reasonable performance where deep learning would overfit [50].

5.4 Identified limitations and research gaps

Table 6 outlines major limitations observed when applying traditional machine learning techniques to ECG-based cardiovascular analysis. It highlights challenges related to feature engineering, temporal signal interpretation, and dataset imbalance, which can affect model reliability and clinical applicability. Understanding these limitations helps guide the development of more robust AI-based diagnostic systems.

5.5 When to choose traditional ML over deep learning

Based on our synthesis, traditional ML remains preferable when:

1. *Dataset size is limited* (< 1000 recordings)
2. *Interpretability is paramount* (clinical decision support requiring explanation)
3. *Computational resources are constrained* (edge devices, real-time applications)
4. *Well-established features exist* (conditions with known ECG correlates)
5. *Rapid prototyping is needed* (quick baseline establishment)

Conversely, deep learning is recommended for large datasets, novel pattern discovery, and end-to-end learning scenarios—as discussed in Sect. 6.

6 Deep learning approaches

Deep learning methods have become the dominant approach for ECG-based cardiovascular disease prediction, appearing in 78% of reviewed studies from 2023–2025. Unlike traditional machine learning, deep learning architectures learn hierarchical representations directly from data, eliminating the need for manual feature engineering. This section examines the principal deep learning architectures identified in our review and their applications to ECG analysis.

6.1 Convolutional neural networks (CNNs)

Convolutional Neural Networks are the most widely used deep learning architecture for ECG analysis, employed in 67% of reviewed studies. CNNs are designed to automatically

Table 6 Key limitations of traditional machine learning in ECG-based cardiovascular analysis

Limitation	Description	Impact
Feature engineering bottleneck	Performance depends heavily on expert-crafted features	May miss novel patterns; limits scalability to new conditions
Temporal dynamics capture	Most ML methods use aggregated features, losing beat-to-beat variations	Reduced sensitivity to rhythm disorders and transient events
Multi-lead integration	Traditional methods struggle to model inter-lead relationships	Loss of spatial information in 12-lead ECGs
Generalization challenges	ML models show greater performance degradation across datasets than deep learning	Average 10–15% accuracy drop in cross-dataset validation
Class imbalance sensitivity	Most ML algorithms assume balanced classes	Poor performance on rare but clinically important conditions

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learn spatial hierarchies of features through convolutional operations that apply learnable filters across input signals [51].

Architecture Overview A typical CNN for ECG classification consists of three component types:

- *Convolutional layers* that apply filters to detect local patterns (e.g., QRS complexes, P waves)
- *Pooling layers* that reduce dimensionality and provide translation invariance
- *Fully connected layers* that map extracted features to classification outputs

For a 1-D ECG signal x = {x_1, x_2, …, x_N}, the convolutional layer operation is defined as:

$$\mathrm{yj} = \mathrm{f}(\sum \mathrm{iKij} \otimes \mathrm{xi} + \mathrm{bj})$$

$$y_j = f(\sum iK_{ij} \otimes x_i + b_j) \tag{1}$$

where y_j represents the output feature map, K_{ij} denotes the learnable kernel, o times is the convolution operation, and b_j is the bias term.

Strengths in ECG analysis

- Excellent at capturing local morphological patterns (QRS morphology, ST segment changes)
- Translation invariance enables detection regardless of exact timing
- Hierarchical feature learning from raw signals without manual engineering

Limitations

- Limited ability to capture long-range temporal dependencies spanning multiple heartbeats
- Performance degrades with significant noise or baseline wander
- Require substantial training data to generalize across patient populations

6.2 Recurrent neural networks and LSTM

Long Short-Term Memory networks, a specialized form of recurrent neural networks, address the limitations of CNNs in capturing temporal dependencies [52]. LSTMs maintain internal memory states that can preserve information across hundreds of time steps, making them particularly suitable for rhythm analysis. Table 7 summarizes the main gates used in Long Short-Term Memory (LSTM) networks and their corresponding mathematical formulations. Each gate plays a specific role in controlling how information is stored, updated, and passed through the sequence during temporal learning.

LSTM Architecture: The LSTM cell regulates information flow through four gates:

Table 7 Functional gates and mathematical formulation in LSTM networks

Gate	Function	Equation
Forget gate f_t	Controls what information to discard from memory	$f_t = \sigma\left(w_f.[h_{t-1}, x_t] + b_f\right)$(2)
Input gate i_t	Determines what new information to store	$i_t = \sigma\left(w_i.[h_{t-1}, x_t] + b_i\right)$(3)
Update gate g_t	Creates candidate memory values	$g_t = tanh\left(w_g.[h_{t-1}, x_t] + b_g\right)$(4)
Output gate o_t	Controls information output to next layer	$o_t = \sigma\left(w_o.[h_{t-1}, x_t] + b_o\right)$(5)

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where h_{t-1} is the previous hidden state, x_t is the current input, w represents weight matrices, and b denotes biases.

Applications in ECG analysis:

- Atrial fibrillation detection (capturing irregular rhythm patterns)
- Arrhythmia classification from long-term recordings
- Predictive modeling using temporal progression
- Computationally intensive for long sequences
- Susceptible to vanishing gradients with very long dependencies
- Less effective at capturing fine-grained morphological details

6.3 Hybrid CNN-LSTM architectures

Hybrid models combining CNNs and LSTMs have emerged as the state-of-the-art for many ECG tasks, appearing in 34% of recent studies. These architectures leverage CNNs for morphological feature extraction and LSTMs for temporal modeling [53].

Typical Architecture

1. CNN layers extract local waveform features from ECG segments
2. Feature sequences are fed to LSTM layers for temporal modeling
3. Fully connected layers produce final classifications

Performance Advantage: Hybrid models consistently outperform standalone CNNs or LSTMs by 3–8% on benchmark datasets, particularly for conditions involving both morphological abnormalities and rhythm disturbances [53, 54].

6.4 Autoencoders

Autoencoders are unsupervised learning architectures that learn compressed representations by reconstructing their input [55]. In ECG analysis, they serve three primary purposes:

Denoising Autoencoders Trained to reconstruct clean signals from noisy inputs, these models learn to preserve diagnostic content while suppressing artifacts [45].

Variational Autoencoders (VAEs) Used for anomaly detection by learning the distribution of normal ECG patterns and identifying deviations [49].

Representation Learning The encoder portion can be fine-tuned for classification tasks, particularly useful when labeled data is limited.

As shown in Eq. (6), the encoder transforms an input vector x into a new vector z.

$$z = h^{(l)} + \left(W^{(l)}x + b^{(l)}\right) \tag{6}$$

where represents the relevant layer, h is the encoder's transfer function, W and b is the weight and bias. The decoder turns the encoded data into an actual data, represented as $\hat{x}$ as shown in Eq. (9).

$$\hat{x} = h^{(l)} + \left(W^{(l)}z + b^{(l)}\right) \tag{7}$$

The encoder and decoder were denoted by the logistic sigmoid function, as defined by the Eq. (8):

$$f_z = \frac{1}{1 + exp^{-z}} \tag{8}$$

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The hidden layers have 10 neurons, and the cost function, which is equivalent to the mean squared error, was utilised as the optimization criterion.

Training minimizes reconstruction error, typically mean squared error between input and output.

6.5 Transfer learning

Transfer learning addresses the limited size of medical datasets by leveraging knowledge from related tasks or domains [56]. In ECG analysis, this typically involves:

Pre-training on large public datasets Models are first trained on extensive ECG repositories (e.g., PTB-XL with 21,799 recordings), then fine-tuned for specific tasks with smaller datasets.

Cross-domain transfer Knowledge from image classification models (trained on ImageNet) is adapted to ECG analysis, though this requires careful 1D-to-2D signal conversion.

Common pre-trained architectures adapted for ECG ResNet50, Inception-v3, DenseNet121, and EfficientNet variants have been successfully applied after appropriate input reshaping.

Benefits:

- Reduces training data requirements by 60–80%
- Accelerates convergence and improves final performance
- Enables deep learning on smaller clinical datasets

6.6 Emerging architectures

Transformers and Attention Mechanisms Self-attention architectures, originally developed for natural language processing, are gaining traction for ECG analysis (15% of 2024–2025 studies). They capture long-range dependencies without the sequential processing limitations of RNNs.

Graph Neural Networks Emerging approaches model multi-lead ECGs as graphs, capturing spatial relationships between leads that CNNs treat independently [57].

6.7 Comparative performance analysis

Table 8 compares several deep learning architectures commonly used for ECG-based cardiovascular disease detection. It highlights their performance across different clinical tasks such as arrhythmia, myocardial infarction, and atrial fibrillation detection. The

Table 8 Performance comparison of deep learning architectures across ECG tasks

Architecture	Arrhythmia detection (Accuracy %)	MI detection (Accuracy %)	AF detection (AUC)	Inference speed	Training data needed
CNN (standalone)	92–96	91–95	0.94–0.97	Fast	High
LSTM/RNN	88–93	85–90	0.91–0.95	Medium	Medium
CNN-LSTM Hybrid	95–98	94–97	0.97–0.99	Medium	High
Autoencoder-based	87–92	86–91	0.89–0.94	Fast	Low (unsupervised)
Transformer	94–97	93–96	0.96–0.98	Slow	Very High
Transfer Learning	93–97	92–96	0.95–0.98	Fast	Low (fine-tuning)

Based on synthesis of studies reporting comparable metrics on standard datasets [46–50, 53, 54, 57–61]

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comparison also includes inference speed and training data requirements to illustrate practical trade-offs between accuracy and computational cost.

6.8 Key limitations and research directions

Table 9 summarizes important challenges encountered when applying deep learning techniques to ECG-based cardiovascular analysis. It also highlights current research directions and technological approaches that aim to address issues such as data dependency, interpretability, and computational efficiency.

7 Explainable AI

XAI is important for getting the trust of clinicians and cardiologists when integrating automated systems into their workflow. XAI methods support this by providing two forms of interpretability: interpretation of results using feature relevance and interpretation of results through visual explanations. Feature relevance-based AI model explanation is a method of evaluating the model's decision. This technique generates a score that indicates how each feature impacts to the trained model's prediction. There are several ways to describe AI models using future relevance. However, this chapter briefly examines SHapley Additive Explanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), and etc.

SHAP

SHAP are depend on game theory, and Shapley score describe each team member's marginal contribution [62]. When analyzing ML models, SHapley scores represent each feature's contribution to a single black box model's classification output d [63]. Given a dataset of sizeNxM, where S is a feature's subset,$F = \{F_1, F_2, ..., F_k, ..., F_M\}$, where F_k represent features of the k-th column. A model f's output can be influenced by feature F_i in two different ways. Initially, feature F_i is employed for model training, and the final model is denoted by$f_{S\cup\{i\}}$. Second, $f_{S\cup\{i\}}$ and f_S are derived with the aid of the previously trained model$f$. The SHapley score (ϕ_i) for the feature F_i is calculated by Eq. (9).

$$\phi_i = \sum_{S\subseteq F\setminus\{i\}} \frac{|S|!\,(|F|-|S|-1)!}{|F|!}\left[f_{S\cup\{i\}}\left(x_{S\cup\{i\}} - f_S\left(x_S\right)\right)\right] \tag{9}$$

where x_S represents the input feature values, $f_S\left(x_S\right)$ represents the f's marginal value.

LIME

LIME explains which attributes have most impact on the decision of the AI black box model by approximating large nonlinear AI models using a locally interpretable surrogate model [64]. Advanced models are assumed to be linear at the local scale in

Table 9 Key challenges and emerging solutions in deep learning-based ECG analysis

Limitation	Description	Current solutions
Data hunger	Deep learning requires large annotated datasets	Transfer learning, data augmentation, synthetic data
Generalization gap	Performance drops 8–15% across datasets	Domain adaptation, multi-center training
Interpretability	"Black box" decisions limit clinical trust	XAI integration (Section VII)
Temporal modeling	Capturing very long dependencies	Transformers, attention mechanisms
Computational cost	Deployment on edge devices	Model pruning, quantization, efficient architectures

Based on synthesis of studies reporting comparable metrics on standard datasets [46–50, 53, 54, 57–61]

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this approximation. Therefore, it could be possible to approximate the complex model in terms of individual events that need to be reported. To measure neighborhood relevance, the penalty function $\pi_x(z)$ evaluates how close perturbed cases, $z \in R$, are to an instance feature vector, x. Given a class of presumably interpretable models G, $g \in G$, and a black box model f, g is the surrogate model that most closely resembles f. Equation (10) demonstrates how to minimize the objective function $L(f, g, \pi_x) + \Omega(g)$ to achieve the explanation $\xi(x)$.

$$\xi(x) = argmax\left(L(f, g, \pi_x) + \Omega(g)\right)_{g \in G} \tag{10}$$

where L represents the loss function that evaluates g's accuracy in approximating f in a given locality (π_x) and complexity $\Omega(g)$.

LIME chooses interpretable representation quality $x' \in \{0,1\}^{d'}$ from the dataset's initial feature space,$x \in X$. The mapping function h_x transforms the binary features into their original real-valued representation, resulting in $h_x : z' \to z$ and z=$h_x\left(z'\right)$. Equation (11) creates the locality-aware loss function using the revised sample dataset,Z, and its labels, i.e.,$\left\{\left(z', f(z)\right)\right\}$.

$$L(f, g, \pi_x) = \sum_{z, z' \in Z} \pi_x(z)\left(f(z) - g\left(z'\right)\right)^2 \tag{11}$$

LIME provides an accessible explanation, but it is dependent on the complexity of the models [65].

Another method for evaluating a black box AI model's prediction is to highlight the data segments that are most essential to the model's judgment. To explain black box AI classifiers in ECG signal-based CVD detection, visual explanation-based result presentation methodologies were comprehensively examined. Examples include saliency maps (SM), and CAM-based approaches, among others.

Class Activation Maps

The CAM technique gives visual details by highlighting the key areas in input that influence ML model decisions [66]. The CAM evaluates how units in the CL's final activation filter map (F_{ij}^k) affect the output layer's class prediction score (y^c) [67]. The CAM technique that combines GAP and FCL to yield L_{ij}^c. Equation (12) describes the linear relationship between F_{ij}^k and y^c. Table 5 gives an overview of the existing ASI methods and their limitations

$$y^c = \sum_k w_k^c \sum_i \sum_j F_{ij}^k \tag{12}$$

The FCL's weight for filter k is w_k^c, c, i, and j represent the feature map's indices, c represents the class, and k represents the filter. CAM seeks to determine the contribution of final feature maps satisfying the equation $y^c = \sum_{i,j} L_{ij}^c$. Equation (12) calculates each unit's contribution to the final feature map, L_{ij}^c, as shown in Eq. (13).

$$L_{ij}^c = \sum_k w_k^c F_{ij}^k \tag{13}$$

Equation (14) provides the CAM for class c in a 1-D time series data like ECG.

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$$L_t^c = \sum_k w_k^c F_t^k \tag{14}$$

where L_t^c represent the significance of the temporal position t's activation, leads to signal classification into class c, and F_t^k denotes the activation of k filter at the temporal instance t. CAM was used to assess a CNN's classification of an ECG signal. As a result, it enable for the observation of which ECG portion that are heavily influenced by the classification model's choice [66].

Saliency Maps

The feature SM shows which signal components are most important in identifying the incoming signal [68]. The SM is created by calculating the gradients between an ML model's output, $y_c(x)$, and its input, x, for the class c. Equation (15) shows how the outcome y_c can be calculated by utilizing the 1st-order Taylor expression.

$$y_c(x) \approx w^T x + b \tag{15}$$

The w defined in Eq. (16) represent the gradient that explains the model classification result.

$$w = \frac{\partial y_c(x)}{\partial x} \tag{16}$$

SMs were used to explain advanced ML models in ECG-based CVD identification. Table 10 gives an overview of XAI techniques applied to ECG classification, highlighting the major strengths and constraints of each method.

8 Evaluation metrics

Researchers employ a wide range of metrics to evaluate the effectiveness of CVD detection algorithms. Accuracy, specificity, recall, F1-score, precision, false negative rate (FNR), and false positive rate (FPR) are some of the most commonly used evaluation measures [75, 76]. These metrics give additional insight into multiple elements of model performance, such as overall reliability, class-specific behavior, and correlation strength. The formulas for computing these measurements are shown in Eqs. (17–23).

Table 10 Overview of existing Xai methods and their limitations

Ref. No	Technique used	Key insights	Research gap
[69]	Comparative Survey	Reviewed XAI evaluation methods in cardiology AI models	Over 40% of ECG studies lack clinician-validated interpretability, limiting clinical adoption
[70]	Ensemble Models + Limited XAI	Evaluated AI models for clinical risk prediction	Non-interpretable models showed inconsistent results, highlighting the need for reliable explainability
[71]	Random Forest + SHAP	Applied SHAP to interpret ECG features for LVEF prediction	Provides global feature importance but lacks local waveform morphology interpretation
[72]	XGBoost + SHAP	Predicted post-MI in-hospital mortality using ECG features	Explanations remain coarse and do not capture waveform-level reasoning for clinicians
[73]	Gradient Boosting + SHAP	Integrated ECG, EEG, and clinical data for neurological outcome prediction	Feature importance lacks segment-level interpretability; ECG contribution remains unclear
[59]	CNN + Explainability Map	Introduced lead-level visualization for CVD detection	Limited clinical validation and insufficient physiological explanation detail
[74]	SVM / Random Forest + SHAP/LIME	Compared interpretability techniques for ECG rhythm classification	High computational cost and inconsistent explanations reduce reliability

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$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

$$Specificity = \frac{TN}{TN + FP} \tag{18}$$

$$Recall = \frac{TP}{TP + FN} \tag{19}$$

$$Precision = \frac{TP}{TP + FP} \tag{20}$$

$$F1 - Score = \frac{TP + TN}{TP + TN + FP + FN} \tag{21}$$

$$FNR = \frac{FN}{TP + FN} \tag{22}$$

$$FPR = \frac{FP}{TN + FP} \tag{23}$$

In the above equations, TP and TN denote True Positives and Negatives, FP and FN denote False Positives and Negatives.

9 Challenges and future direction

The systematic review of 83 high-quality studies reveals that despite significant advances in AI-based ECG analysis for cardiovascular disease detection, several persistent challenges continue to limit clinical translation and real-world applicability. These challenges span the entire development pipeline—from data acquisition through model deployment—and require coordinated interdisciplinary efforts to address. This section synthesizes the key gaps identified across the literature and proposes concrete future research directions.

9.1 Data-related challenges

9.1.1 Class imbalance and dataset bias

The majority of publicly available ECG datasets exhibit substantial class imbalance, with normal sinus rhythms vastly outnumbering pathological recordings. This imbalance skews model learning toward majority classes, leaving minority conditions—often those most clinically urgent—poorly represented. Standard augmentation techniques and class-weighted loss functions provide partial remedies but do not fully address the underlying problem of insufficient pathological exemplars.

9.1.2 Noise and artifact contamination

Raw ECG signals are frequently corrupted by baseline wander, powerline interference, motion artifacts, and muscular activity. While preprocessing techniques including wavelet transforms, empirical mode decomposition, and adaptive filtering can mitigate these distortions, they introduce their own challenges: increased computational overhead, potential suppression of subtle morphological features, and parameter sensitivity that varies across datasets and recording conditions.

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9.1.3 Annotation quality and label noise

Manual annotation by human experts remains the gold standard for establishing ground truth, yet inter-annotator variability and occasional errors introduce label noise that propagates through training. Few studies report inter-annotator agreement metrics or employ strategies to detect and correct annotation inconsistencies. This lack of attention to label quality undermines the reliability of supervised learning approaches.

9.1.4 Demographic diversity gaps

As noted in Sect. 3.4, 78% of public datasets originate from North America or Europe, with minimal representation of pediatric populations, older adults, and diverse ethnic groups. Models trained on such homogeneous data may underperform when deployed in settings serving different populations, raising concerns about algorithmic fairness and healthcare equity.

9.1.5 Future research directions for data challenges

Table 11 outlines major research challenges identified in current ECG-based AI studies and highlights promising directions for future investigation. The proposed strategies aim to improve data quality, model robustness, fairness, and scalability in real-world cardiovascular applications.

9.2 Model design and performance limitations

9.2.1 Trade-offs between complexity and interpretability

Traditional machine learning approaches offer computational efficiency and relative interpretability but depend heavily on hand-crafted features that may not capture the full complexity of ECG morphology. Deep learning alternatives learn rich representations directly from data yet function as opaque "black boxes" that resist interpretation. This trade-off between performance and transparency remains unresolved.

9.2.2 Generalization across populations and settings

Our analysis reveals that only 28% of studies include external validation, and cross-dataset performance typically degrades by 8–15% compared to within-dataset evaluation. Models trained on single-center data often fail to generalize to different patient populations, recording equipment, or clinical settings—a critical barrier to widespread adoption.

Table 11 Key research challenges and future directions in AI-based ECG analysis

Challenge	Proposed research directions
Class imbalance	Develop synthetic data generation techniques (e.g., GAN-based augmentation) that preserve pathological realism; explore few-shot learning and meta-learning approaches
Noise robustness	Create noise-resistant architectures that learn directly from minimally preprocessed signals; develop standardized noise benchmarks for fair comparison
Annotation quality	Implement uncertainty-aware learning methods that account for label noise; develop semi-supervised and active learning approaches to reduce annotation burden
Demographic diversity	Establish multi-center, multinational data collection consortia; mandate demographic reporting in public datasets; evaluate model performance across subgroups

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9.2.3 Temporal dynamics and inter-lead interactions

Current architectures, even hybrid CNN-LSTM models, struggle to capture long-range temporal dependencies spanning hundreds of heartbeats. Similarly, inter-lead interactions in multi-lead ECGs remain underexplored, with most studies treating leads independently rather than modeling their relationships.

9.2.4 Computational constraints for deployment

Deep models with millions of parameters exceed the computational capabilities of wearable devices and point-of-care systems. Model compression, quantization, and efficient architecture design are essential for real-time deployment but receive limited attention in the literature.

9.2.5 Future research directions for model development

Table 12 highlights important challenges that affect the real-world deployment of AI models for ECG analysis. It also outlines potential research directions aimed at improving interpretability, generalization across populations, temporal signal modeling, and computational efficiency for practical clinical use.

9.3 Technological and methodological constraints

9.3.1 Reproducibility crisis

The field faces a reproducibility challenge: only 29% of reviewed studies made code publicly available, and preprocessing details were often insufficiently documented to enable replication. This lack of transparency hinders scientific progress and clinical translation.

9.3.2 Evaluation inconsistencies

Studies employ diverse evaluation metrics, validation strategies, and reporting standards, making cross-study comparison difficult. The absence of standardized benchmarks for specific clinical tasks prevents meaningful assessment of progress over time.

9.3.3 Ablation studies and component analysis

Few studies include comprehensive ablation analyses that quantify the contribution of individual preprocessing steps, architectural components, or hyperparameter choices. This limits understanding of what drives model performance and impedes principled design improvement.

Table 12 Emerging challenges and strategic research directions for deployable ECG AI systems

Challenge	Proposed research directions
Interpretability-performance trade-off	Develop inherently interpretable architectures (e.g., attention-based models that highlight relevant waveform regions); integrate physiological knowledge into model design
Generalization	Mandate external validation as a publication requirement; develop domain adaptation techniques that adjust models to new populations with minimal fine-tuning
Temporal modeling	Explore transformer architectures and graph neural networks for capturing long-range dependencies and lead interactions
Deployment efficiency	Investigate model pruning, knowledge distillation, and neural architecture search for lightweight models suitable for edge deployment

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9.3.4 Future research directions for methodological rigor

Table 13 presents key methodological challenges that affect the reliability and transparency of ECG-based AI research. It also highlights recommended practices such as standardized evaluation protocols, reproducibility guidelines, and systematic model analysis to improve research quality and comparability.

9.4 Clinical integration and wider implications

9.4.1 Gap between research and clinical reality

Most published models are evaluated under idealized conditions that poorly reflect clinical practice. Factors such as variable signal quality, missing leads, concurrent medications, and multiple comorbidities are rarely addressed, yet they profoundly impact real-world performance.

9.4.2 Interpretability for clinical trust

Current explainability methods, while technically sophisticated, often fail to provide clinically meaningful insights. Saliency maps highlighting signal regions may satisfy technical curiosity but do not answer the questions clinicians actually ask: "Does this model's reasoning align with established pathophysiology?

Should I trust this recommendation for my specific patient?".

9.4.3 Ethical and regulatory challenges

Training data bias can propagate into diagnostic disparities across demographic groups, yet fairness assessments remain rare. Data privacy concerns limit data sharing, while regulatory pathways for AI-based medical devices remain uncertain and vary across jurisdictions.

9.4.4 Workforce and implementation considerations

Successful clinical integration requires not only technical validity but also workflow compatibility, user training, and strategies for handling model failures. These human factors receive minimal attention in the technical literature.

9.4.5 Future research directions for clinical translation

Table 14 highlights key clinical, ethical, and regulatory challenges associated with integrating AI-driven ECG analysis into healthcare practice. It also outlines research directions aimed at improving clinical validity, interpretability, fairness, and safe adoption in real-world medical environments.

Table 13 Methodological challenges and best practices for reproducible ECG AI research

Challenge	Proposed research directions
Reproducibility	Establish community standards for code and data sharing; require detailed preprocessing and training documentation
Standardized evaluation	Develop task-specific benchmark suites with consistent metrics and validation protocols; create living leaderboards for continuous assessment
Ablation studies	Encourage systematic component analysis in publications; develop automated tools for sensitivity analysis

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Table 14 Clinical and ethical challenges in deploying AI for ECG-based cardiovascular care

Challenge	Proposed research directions
Clinical realism	Evaluate models on data reflecting real-world conditions; incorporate point-of-care constraints into study design
Meaningful interpretability	Develop explanation methods that connect model decisions to physiological concepts; validate explanations with clinician users
Fairness and ethics	Mandate demographic subgroup analysis; develop debiasing techniques; establish ethical guidelines for AI deployment
Regulatory pathways	Engage with regulatory agencies early; develop validation frameworks that satisfy approval requirements
Implementation science	Study human-AI interaction in clinical settings; develop training programs and failure management protocols

9.5 Synthesis and roadmap

The challenges identified across these domains are interconnected. Data diversity limitations exacerbate generalization problems. Poor interpretability compounds clinical trust barriers. Methodological inconsistencies slow progress on all fronts. Addressing these challenges requires coordinated action:

Near-term priorities (1–3 years):

- Establish standardized evaluation benchmarks for common clinical tasks
- Mandate demographic reporting and external validation in publications
- Develop and disseminate reproducible preprocessing pipelines

Medium-term priorities (3–5 years):

- Create multi-center, demographically diverse data collection consortia
- Integrate explainability methods with clinical workflow requirements
- Develop regulatory approval pathways through early engagement

Long-term priorities (5+ years):

- Achieve prospective clinical validation in diverse practice settings
- Establish continuous learning systems that adapt to population changes
- Realize equitable, trustworthy AI-assisted cardiovascular care

The path from research innovation to clinical impact requires not merely technical advances but fundamental shifts in how the field approaches problem formulation, evaluation, and collaboration. Engineers must work alongside clinicians from project inception. Researchers must prioritize generalizability over benchmark performance. The community must embrace transparency and reproducibility as core values. With such commitment, the promise of AI-enhanced cardiovascular care can be realized.

10 Conclusion

10.1 Synthesis of key findings

This systematic review has examined the landscape of artificial intelligence applications for ECG-based cardiovascular disease prediction, analyzing 83 high-quality studies published between 2020 and 2025. Our investigation spanned the entire development pipeline—from data acquisition and preprocessing through model architecture selection, explainability integration, and performance evaluation. The evidence gathered reveals both remarkable progress and persistent challenges that shape the field's trajectory.

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Deep learning approaches, particularly convolutional neural networks and hybrid architectures combining CNN with LSTM components, have demonstrated impressive technical capabilities. When evaluated on benchmark datasets under controlled conditions, these models achieve accuracy rates exceeding 95% for arrhythmia classification and show promise for more complex tasks like myocardial infarction detection and heart failure risk stratification. The field has matured beyond simple proof-of-concept demonstrations toward more sophisticated applications that incorporate attention mechanisms, transfer learning, and multimodal data integration.

10.2 Persistent challenges and critical gaps

Despite these advances, our analysis reveals that significant obstacles continue to hinder clinical translation. The methodological rigor of published research varies considerably—only 38.6% of reviewed studies demonstrated low risk of bias according to our quality assessment framework. This finding raises important questions about the reproducibility and reliability of reported results.

Data-related challenges remain paramount. Class imbalance continues to skew model learning toward majority conditions while underrepresenting clinically important but less frequent pathologies. Noise artifacts from motion, electrode displacement, and powerline interference corrupt signal quality, and while preprocessing techniques can mitigate these effects, they may also inadvertently remove diagnostically relevant information. Most concerning is the limited demographic diversity across datasets—78% originate from North America or Europe, with minimal representation of pediatric populations, older adults, and diverse ethnic groups. Models trained on such homogeneous data risk performing poorly when deployed in real-world settings serving varied populations.

Methodological limitations further compound these data problems. The field lacks standardized evaluation protocols, making cross-study comparisons difficult. External validation—testing models on data independent from training sources—appears in only 28% of studies, yet this practice is essential for assessing generalizability. When researchers do report performance metrics, they often emphasize overall accuracy while neglecting class-specific performance, calibration, and uncertainty quantification—factors that matter greatly in clinical decision-making.

Interpretability deficits create additional barriers. While 52% of recent studies incorporate some form of explainable AI, the clinical utility of these explanations remains questionable. Saliency maps and feature importance scores may satisfy technical curiosity but often fail to provide the physiological insights clinicians need to trust and act upon model recommendations.

10.3 Toward clinical translation

Bridging the gap between research prototypes and clinically deployed systems requires fundamental shifts in how the field approaches AI development. Our analysis suggests several priority directions:

Data infrastructure must expand beyond existing repositories to include diverse, multinational, longitudinally tracked populations. New datasets should systematically capture demographic variables, comorbidity information, and annotation confidence. Federated learning approaches offer promise for leveraging distributed data while

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preserving privacy, enabling model training across institutions without centralizing sensitive information.

Methodological standards require urgent attention. The research community should work toward consensus on evaluation protocols that include mandatory external validation, standardized reporting of performance across demographic subgroups, and assessment of model calibration and uncertainty. Open-source code and pre-trained model sharing would accelerate progress while enhancing reproducibility.

Interpretability research must refocus from technical explanation toward clinical communication. Effective XAI should not merely highlight which signal regions influenced predictions but connect these patterns to established physiological knowledge. Collaborations between AI researchers and clinician-scientists are essential for developing explanation frameworks that support rather than supplant human judgment.

Regulatory and ethical frameworks need parallel development. Models destined for clinical use must undergo rigorous validation not just for accuracy but for fairness across populations, robustness to distribution shifts, and transparency in failure modes. The research community should engage proactively with regulatory agencies to establish clear pathways from innovation to approved medical devices.

10.4 The path forward

The integration of artificial intelligence into cardiovascular care holds genuine promise for improving patient outcomes, reducing healthcare disparities, and supporting overburdened clinical systems. Realizing this promise requires moving beyond the current paradigm where each research group pursues isolated innovations toward coordinated, collaborative efforts that pool expertise and resources.

Engineers must work alongside clinicians from the earliest stages of model development, ensuring that problem formulations reflect genuine clinical needs and that evaluation metrics capture outcomes that matter to patients and providers. Clinicians must engage with the technical complexities of AI systems, developing enough fluency to critically evaluate model outputs and identify potential failures. Regulators and ethicists must join these conversations early, helping shape development practices that prioritize safety, fairness, and transparency from the outset.

10.5 Final reflections

This review has attempted to provide not merely a catalog of existing work but a critical analysis that identifies both achievements and shortcomings. For researchers entering this field, we hope it serves as a practical roadmap—highlighting established techniques, flagging methodological pitfalls, and pointing toward unanswered questions that merit investigation. For experienced investigators, we hope it stimulates reflection on how individual research efforts contribute to the collective goal of improving cardiovascular care.

The convergence of abundant ECG data, powerful computational methods, and pressing clinical needs creates an unprecedented opportunity. Whether this opportunity translates into meaningful advances depends on whether the research community can address the gaps identified here with rigor, creativity, and commitment to the ultimate purpose of this work: helping patients live longer, healthier lives.

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Author contributions
Thanuja M: Conceptualization of the review scope and objectives, development of the search strategy and methodology, conducting the systematic literature search and screening, data extraction and curation, quality assessment of included studies, writing the original draft of the manuscript. **Soumyashree M Panchal:** Supervision of the entire research process and validation of the methodology. **Prasanna Lakshmi G.S:** Verified And Validated the review process. All authors read and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval
This study is a systematic review of previously published literature and did not involve direct research on human participants, animal subjects, or collection of primary data. All analyzed data were obtained from publicly available sources and previously published studies that had obtained appropriate ethical approvals from their respective institutional review boards. Therefore, no new ethical approval was required for this review.

Consent to participate
Not applicable. This systematic review did not involve direct participation of human subjects. All included studies utilized data from publicly available repositories or previously published research where informed consent was obtained by the original investigators in accordance with the Declaration of Helsinki and applicable local regulations.

Consent for publication
Not applicable. This manuscript does not contain any individual person's data in any form, including individual details, images, videos, or identifiable information. All figures and tables are original creations by the authors based on synthesized information from reviewed literature and do not reproduce copyrighted material from other sources.

Competing interests
The authors declare no competing interests.

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